



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th Dec 2025	CIE - II	Max. Marks : 10 + 50
Semester : V	UG	Duration : 30 + 90 Min
Course Title: ENTREPRENEURSHIP & INTELLECTUAL PROPERTY RIGHTS		Course Code : HS351TA

Part A

Sl. No	Questions	M	BT	CO
1.	Distinguish between Proof of Concept (PoC) and Prototype with respect to risk mitigation.	2	1	3
2.	Why is technical feasibility meaningless without market feasibility? Justify.	2	1	3
3.	How does reverse engineering weaken trade secret protection?	2	2	3
4.	Why is novelty mandatory but aesthetic appeal critical in industrial design protection?	2	3	3
5.	Define Copyright.	2	2	3

Part B

Sl. No	Questions	M	BT	CO
1.	Explain the elements of a business plan. Describe each component with suitable examples.	10	2	2
2.	Discuss the complete Entrepreneurial Opportunity Evaluation Process covering: • Identification of market opportunities and trends • Technical feasibility (Prototype and Proof of Concept) • Financial feasibility (Cost estimation, revenue projection, and break-even analysis) • Role of cross-disciplinary collaboration in technological innovation	10	3	3
3.	Explain the acts which do not constitute Infringement under the Copyright Law.	10	2	3
4.	Explain in detail the complete procedure for obtaining Industrial Design protection in India. Further, critically evaluate: • Grounds and procedure for revocation of registered designs • Types of design infringement • Civil and criminal remedies available Illustrate your answer with at least one real case study.	10	3	2
5.	A software company develops a unique algorithm for data security and keeps it confidential. A former employee illegally discloses this information to a competitor. 1. Identify the type of intellectual property right involved. 2. Explain the legal remedies available under Indian law. 3. Discuss the importance of trade secret protection with reference to this case.	10	3	3



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INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15/12/2025	CIE – II-A	Max. Marks : 10 + 50
Semester : V	UG	Duration : 30 + 90 Min
Course Title: Operations Management		Course Code : IM352IA

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
Part – A				
1.	In time series models, regular fluctuations occurring within a year due to weather or festivals are called _____ variations.	1	1	1
2.	The difference between the actual value and the forecasted value is known as the _____.	1	1	2
3.	Qualitative forecasting methods rely heavily on _____ judgment.	1	1	1
4.	A hotel predicts weekend occupancy using past booking patterns. This uses a _____ time series model.	1	2	1
5.	A company uses a senior manager's opinion to forecast sales of a new product. This reflects the _____ forecasting method.	1	1	1
6.	A production line experiencing a delay at one critical machine identifies it as the _____ resource.	1	2	4
7.	A production manager selects the job with the shortest processing time to reduce average waiting time. This applies the _____ rule.	1	2	3
8.	A plant with multiple machines generates several two-machine equivalent sequences to approximate the optimal schedule. This procedure follows the _____.	1	2	3
9.	A manufacturer with two machines creates an optimal job sequence by choosing the smallest processing time among both machines. This applies _____ rule.	1	2	3
10.	For two jobs processed on multiple machines, the job with smaller processing time on the first machine is scheduled _____ in the Johnson-like rule for job shops.	1	2	3
Part – B				
1	Explain the importance of measuring forecast error in Operations Management. Discuss the managerial implications of each measure with suitable examples.	08	2	3
2	With suitable example, differentiate between forward scheduling and backward scheduling	08	2	4



3	A lathe manufacturer had noted the sales invoices from January to August and noticed the sales during each month as 10, 11, 10, 11, 14, 15, 14 & 15 units respectively. What would be his estimate of sales for September? (i) If he has followed 4 month weighted moving average (with weights 0.4, 0.3, 0.2 & 0.1). (ii) If he has followed SES with $\alpha = 0.2$ and his estimated sales for January is 11.	12	4	3																										
4	Consider the following single machine scheduling problem. <table><tr><th>Job, j</th><th>Processing time, t_j</th><th>Due date, d_j</th></tr><tr><td>1</td><td>12</td><td>17</td></tr><tr><td>2</td><td>10</td><td>14</td></tr><tr><td>3</td><td>10</td><td>12</td></tr><tr><td>4</td><td>9</td><td>20</td></tr><tr><td>5</td><td>14</td><td>18</td></tr><tr><td>6</td><td>17</td><td>27</td></tr></table> Determine the sequence which will minimize the maximum lateness (L_{max}). Also, determine the L_{max} with respect to the optimal sequence.	Job, j	Processing time, t_j	Due date, d_j	1	12	17	2	10	14	3	10	12	4	9	20	5	14	18	6	17	27	10	4	4					
Job, j	Processing time, t_j	Due date, d_j																												
1	12	17																												
2	10	14																												
3	10	12																												
4	9	20																												
5	14	18																												
6	17	27																												
5	Use graphical method to minimize the time needed to process the following jobs on the machines shown (i.e. for each machine, find the job which should be scheduled first). Also, calculate the total time elapsed to complete both jobs. <table><tr><td rowspan="2">Job 1</td><td>Sequence</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>Time (Hrs).</td><td>3</td><td>4</td><td>2</td><td>6</td><td>2</td></tr><tr><td rowspan="2">Job 2</td><td>Sequence</td><td>B</td><td>C</td><td>A</td><td>D</td><td>E</td></tr><tr><td>Time (Hrs)</td><td>5</td><td>4</td><td>3</td><td>2</td><td>6</td></tr></table>	Job 1	Sequence	A	B	C	D	E	Time (Hrs).	3	4	2	6	2	Job 2	Sequence	B	C	A	D	E	Time (Hrs)	5	4	3	2	6	12	4	4
Job 1	Sequence		A	B	C	D	E																							
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Job 2	Sequence	B	C	A	D	E																								
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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 16th Dec 2025	CIE-II	Max. Marks : 10 + 50
Semester : V	UG	Duration : 30 + 90 Min
Course Title: Quality Assurance		Course Code : IM353IA

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO																											
Part – A																															
1.	Explain the purpose of a p-chart.	2	3	01																											
2.	What is AOQ?	2	2	01																											
3.	State the difference between single and double sampling plans.	2	2	02																											
4.	What is AOQL?	2	3	02																											
5.	What are multiple sampling plans?	2	2	02																											
6.																															
7.	Draw the type - A OC curve for the single sampling plan: $n = 50, C = 1, N = 200$.	10	2	3																											
2	Differentiate between C chart and U chart	10	3	3																											
3	Lots of Bushing screws from a single supplier were inspected 100 percent with the following results. <table><tr><td>Lot No.</td><td>Amount in lot</td><td>Number rejected</td></tr><tr><td>01</td><td>6,000</td><td>183</td></tr><tr><td>02</td><td>6,000</td><td>131</td></tr><tr><td>03</td><td>10,000</td><td>184</td></tr><tr><td>04</td><td>1,783</td><td>23</td></tr><tr><td>05</td><td>3,089</td><td>171</td></tr><tr><td>06</td><td>5,774</td><td>65</td></tr><tr><td>07</td><td>3,000</td><td>46</td></tr><tr><td>08</td><td>2,724</td><td>85</td></tr></table> Draw an appropriate control chart for the data. What can you conclude on the state of control?	Lot No.	Amount in lot	Number rejected	01	6,000	183	02	6,000	131	03	10,000	184	04	1,783	23	05	3,089	171	06	5,774	65	07	3,000	46	08	2,724	85	10	4	3
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08	2,724	85																													



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4	The following table gives the number of missing rivets at the aircraft final inspection:					10	4	4
	Sl No	Air craft No.	Number of missing rivets	Sl No	Air craft No.	Number of missing rivets		
	1	202	07	6	207	11		
	2	203	12	7	208	06		
	3	204	13	8	209	19		
	4	205	18	9	210	04		
	5	206	17	10	211	26		
	Establish an appropriate control chart for the purpose.							
5	Explain the meaning of the following terms:					10	3	4
	i.	Producer's risk	ii.	AQL				
	iii.	Consumer's risk	iv.	LTPD				



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**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date : 16th Dec 2025	CIE - II	Max. Marks : 10 + 50
Semester : V	UG	Duration : 30 + 90 Min
Course Title: FINANCIAL ACCOUNTING & COSTING		Course Code : IM254TA

Note: Answer all the Questions.

Sl. No	Questions	M	BT	CO																						
PART - A																										
1.1	What is meant by Overheads in Costing?	2	1	2																						
1.2	Differentiate between Fixed Costs and Variable Costs.	2	3	2																						
1.3	Identify two sectors for which activity-based costing is suitable.	2	3	4																						
1.4	Comment on how cost accumulation happens in Process Costing with an example.	2	3	4																						
1.5	Job costing is for _____, _____ jobs, while batch costing is for <i>a group of</i> _____ <i>products</i> treated as one batch.	2	2	3																						
PART - B																										
2.a	Enumerate the objectives of Cost Accounting.	05	2	3																						
2.b	Discuss the various elements of Costing.	05	2	3																						
3	Prepare Trial Balance for the following Balances as on 31 st March 2023	10	3	2																						
	<table><tr><th>Balances</th><th>₹</th></tr><tr><td>Cost of Goods Sold</td><td>5,20,000</td></tr><tr><td>Opening Stock</td><td>50,000</td></tr><tr><td>Closing Stock</td><td>50,000</td></tr><tr><td>Salary and Wages</td><td>50,000</td></tr><tr><td>Sales</td><td>8,00,000</td></tr><tr><td>Plant & Machinery</td><td>2,00,000</td></tr><tr><td>Drawing</td><td>50,000</td></tr><tr><td>Investment</td><td>4,30,000</td></tr><tr><td>Creditors</td><td>1,00,000</td></tr><tr><td>Capital</td><td>4,00,000</td></tr></table>	Balances	₹	Cost of Goods Sold	5,20,000	Opening Stock	50,000	Closing Stock	50,000	Salary and Wages	50,000	Sales	8,00,000	Plant & Machinery	2,00,000	Drawing	50,000	Investment	4,30,000	Creditors	1,00,000	Capital	4,00,000			
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4.	Prepare Profit and Loss account from the following particulars for the year ended March 31, 2023: Gross Profit ₹ 4,69,000 Subletting ₹ 10,000 Commission Received ₹ 40,000 Rent, tax, and Insurance ₹ 10,000 Discount Received ₹ 22,000 Export Duty ₹ 15,000 Rent Received ₹ 23,000 Audit Fees ₹ 5,000 Bank Charges ₹ 8,000 Packing Charges(Show Room) ₹ 12,000 Dividend Received ₹ 15,000 Depreciation ₹ 24,000 Administration Expenses ₹ 35,000 Selling Expenses ₹ 25,000 Printing and Stationery ₹ 15,000 Distribution Expenses ₹ 28,000 Office Expenses ₹ 32,000 Trade Expenses ₹ 50,000 General Expenses ₹ 25,000 Carriage outward ₹ 15,000 Legal Expenses ₹ 5,000 Bad Debts ₹ 10,000	10	4	3
5	Write short notes on any three types of costing methods used in organizations.	10	2	3
6	In a manufacturing unit, raw material passes through four processes, I, II, III, and IV and the output of each process is the input for the subsequent process. The losses in the four processes are respectively 25%, 20%, 20% and 16 2/3 % respectively for I, II, III and IV processes of the input. If the end product at the end of the IV process is 40,000 kg, what is the quantity of raw material required to be fed at the beginning of Process I and the cost of the same at Rs. 5 per kg?	10	3	3



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Date : 17/12/2025	CIE – II	Max. Marks : 10 + 50
Semester : V	UG	Duration : 30 + 90 Min
Course Title: Enterprise Information Systems		Course Code : IM355TBB

Note:

1. Answer all questions

Sl. No	Questions	M	BT	CO
Part – A				
1.	Define Robotic Process Automation (RPA) and state its primary focus.	2	1	CO1
2.	State the fundamental reason why a company must engage in Business Process Reengineering (BPR) before implementing any major automation initiative.	2	2	CO2
3.	Define the term Data Mart and state how it differs in scope from a full Data Warehouse.	2	1	CO3
4.	Name the two foundational processes that are typically executed by Data Mining tools to discover new knowledge within a BI system.	2	2	CO3
5.	A decision-maker uses a BI tool to analyze sales by clicking on a region, then a city, and finally a specific store. Name this OLAP technique and briefly state its purpose.	2	2	CO3
Part – B				
1	List and explain various methods of business process automation.	10	2	CO2
2	Define Robotic Process Automation (RPA). Explain how it works. List the uses and benefits.	10	2	CO2
3	Explain the core difference between Business Intelligence (BI) and Data Warehousing. Describe why BI is considered “the technology and practice of applying information to make decisions”.	10	2	CO1
4	Describe the four progressive types of Data Analytics used in an EIS. For each type, state the central question it is designed to answer.	10	3	CO3
5	Apply the concept of Online Analytical Processing (OLAP) by explaining its primary function. Illustrate how a retail manager would use an OLAP technique like Drilling Down to analyze a regional sales report.	10	4	CO3



RV COLLEGE OF ENGINEERING®, Bengaluru-59

(Autonomous Institution Affiliated to VTU, Belagavi)

Program	CHEMICAL ENGINEERING			
Course Title	GREEN AND HYDROGEN TECHNOLOGY			
Course Code	CH375TGD			
CONTINUOUS INTERNAL EVALUATION - 2				
GLOBAL ELECTIVE: GROUP D				
Sem	V	Date Time	DEC 2025	
Max Marks	60	16.12.2025 13:20 - 15:20	Max Duration	120 Min

Instructions to Candidates:

All the questions are compulsory. No additional booklet will be provided.

Q. No.	QUESTIONS	M	BT	CO
1.1	What are the outcomes of hydrogen risk assessment?	02	01	01
1.2	List the steps involved in the quantitative hydrogen risk assessment.	02	01	04
1.3	Define Codes of Practice.	02	01	03
1.4	What is PAS?	02	01	02
1.5	List the parameters associated with hydrogen for which the codes and standards are defined. Hint: The approval for these parameters vary between countries.	02	01	02
Part B				
2	Explain in detail, how hydrogen gas is stored as compressed gas including its advantages and disadvantages.	10	02	04
3	What are the salient features of storing hydrogen as cryogenic liquid? How is it done and what are the associated benefits and hurdles?	10	02	04
4	Is storing hydrogen as a metal hydride a boon or a bane (annoyance)? Justify with substantial details including the working principle, advantages and disadvantages.	10	03	03
5	How are liquid organic hydrogen carriers beneficial in case of storage and transport of hydrogen? Provide a detailed narration.	10	02	03
6	Give an explanatory note on the codes and standards (national and international) associated with the handling and safety of hydrogen gas.	10	03	01

Course Outcomes

At the end of the course the student will be able to,	
1	Understand the importance of hydrogen and its use as an energy carrier
2	Explain the production, storage and handling of hydrogen
3	Analyze the need for hydrogen as an alternate fuel and the associated challenges
4	Appraise the importance of safety, regulations and codes

Cos/BTL	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks	12	04	22	22	10	30	20	.	.	.